

Trading Chart Analyzer

Product Requirements Document

"We explain. You decide."

Confidential · Internal Use Only
© 2026 Trading Chart Analyzer

Table of Contents

1. Executive Summary
2. The Problem
3. The Solution
4. Target Audience
5. User Flow
6. Features
7. Technical Stack
8. Monorepo Structure
9. Data Model
10. AI Analysis Structure
11. Full Data Flow
12. Security & Compliance
13. Regulatory Risk Management
14. Business Model
15. KPIs
16. MVP Timeline
17. Open Questions

1. Executive Summary

A mobile application (React Native) that allows users to upload a screenshot of a stock, crypto, or forex chart and receive an in-depth AI analysis — without buy/sell recommendations — but with a clear educational explanation of what is happening in the chart.

■ **Core Positioning Constraint**

This product is positioned as an educational tool, not a financial advisory service. Every design, copy, and AI-output decision must reinforce that distinction.

2. The Problem

Gap	Description
Data gap	Most retail traders lose money because they don't understand what the chart is telling them.
Tool gap	Existing tools (TradingView, Plus500) display the data — but don't explain it.
Access gap	Human advisory is expensive, not available 24/7, and not scalable for retail users.
Trust gap	Most AI tools of this kind promise profits → unreliable and legally problematic.

3. The Solution

Step	Description
Upload	User picks a chart screenshot from gallery or camera.
Analyze	AI (Claude Vision) processes the image and returns structured JSON.
Explain	App renders a structured educational breakdown of the chart.
Decide	User reads the analysis and makes their own informed decision.

4. Target Audience

Segment	Description	Key Need
Beginners	Just entered trading, don't understand charts	Plain-English explanations of chart patterns
Intermediate	Trading but lacking confidence in chart reading	Validation and deeper indicator breakdown
Crypto Natives	Active 24/7 in crypto markets	Fast, always-on analysis across any asset

5. User Flow

#	Screen	Description
1	App Entry / Onboarding	First-time users see value proposition and educational disclaimer.
2	Authentication	Register or login via email + password. JWT session created.
3	Home Screen	Upload new chart or browse analysis history.
4	Image Selection	Pick from gallery or capture via camera. Supported: PNG, JPG, WebP.
5	Preview Screen	Review image before submission. Option to retake or reselect.
6	Loading Screen	Progress indicator shown while AI processes the image.
7	Analysis Result Screen	Structured result: Trend · Key Levels · Signals · Points to Watch · Disclaimer.
8	Post-Analysis Actions	Save to history, share as image/link, or start a new analysis.

6. Features

6.1 MVP Features

Chart Upload

- Drag & drop or file picker (gallery or camera)
- Supported formats: PNG, JPG, WebP
- Maximum file size: 10 MB
- Preview screen before submission
- Client-side format and size validation

AI Analysis Engine

The AI returns a structured analysis across five categories:

Category	Inputs	Output
Trend	bullish / bearish / sideways	Explains why the trend is what it is
Key Levels	Support & resistance	Price levels with context and reasoning
Technical Signals	RSI, MACD, MAs, Volume	What each indicator shows and means
Points to Watch	Risk factors, events	What to monitor before any decision
Disclaimer	Auto-appended, non-removable	Educational-only notice on every analysis

■ Hard Rule — AI Output Constraint

The AI never uses the words buy, sell, or promises any profit. All framing is: 'here is what the chart shows' — never 'here is what you should do'.

Analysis History

- View past analyses with chart thumbnails
- Up to 50 saved analyses per user
- Filterable by date

Export

- Download analysis as PDF (Pro tier only)
- Share as image or link

Authentication

- Email + password registration
- JWT-based session management
- Password reset flow

6.2 Phase 2 Features

Feature	Value
Multi-timeframe analysis	Upload D1 + H4 + H1 together for richer context
Chart Q&A;	Follow-up chat with the AI about a specific chart
Comparative analysis	Compare two charts side by side
Watchlist	Track assets the user cares about
Push notifications	Alerts based on chart patterns
Team collaboration	Share analyses within a team workspace

7. Technical Stack

■ Core Engineering Principle

no-any rule: any is strictly forbidden across the entire codebase. Every type must be explicitly defined. Enforced via ESLint: @typescript-eslint/no-explicit-any: "error"

7.1 Frontend

Technology	Version	Role
React Native	0.85.3	Mobile framework (iOS + Android)
Expo SDK	55.0.23	Dev tooling, OTA updates, camera/gallery access
NativeWind	4.2.3	Tailwind CSS syntax for React Native styling
React Navigation	7.2.2	Screen navigation
React Query	5.100.9	Server state, caching, loading/error states
Zustand	5.0.13	Global client state management
Zod	4.4.3	Client-side validation and type inference
TypeScript	5.x	Strict mode, no any
Expo Image Picker	55.0.20	Gallery and camera access
Expo File System	55.0.19	Image to base64 conversion

7.2 Backend

Technology	Version	Role
NestJS	11.1.19	Modular Node.js framework
TypeScript	5.x	Strict mode, no any
nestjs-zod	5.3.0	DTO validation with Zod schemas
Drizzle ORM	0.45.2	Type-safe database ORM
PostgreSQL	16	Primary database
AWS S3 SDK v3	3.1044.0	Image storage
Anthropic SDK	0.95.0	Claude Vision API integration
Passport + JWT	11.0.5 / 11.0.2	Authentication
Stripe SDK	22.1.1	Payment processing
Helmet + Throttler	8.1.0 / 6.5.0	Security layer

7.3 Monorepo & DevOps

Technology	Version	Role
NX	22.7.1	Shared package management across apps
Docker + Docker Compose	latest	Local dev environment
GitHub Actions	latest	CI/CD pipeline
ESLint	10.3.0	Code quality
@typescript-eslint	8.59.2	no-explicit-any as error
Prettier	3.8.3	Code formatting
Husky	9.1.7	Pre-commit hooks
lint-staged	17.0.2	Pre-commit hooks

8. Monorepo Structure

```

root/ apps/ mobile/ ← React Native + Expo api/ ← NestJS packages/ types/ ← Shared TypeScript types
(AnalysisResponse, User ...) validators/ ← Shared Zod schemas constants/ ← DISCLAIMER_TEXT, LIMITS,
SYSTEM_PROMPT nx.json tsconfig.base.json .eslintrc.json

```

9. Data Model

9.1 Users Table

Field	Type	Notes
id	UUID	Primary key
email	string	Unique
passwordHash	string	Bcrypt
tier	enum	free / pro / team
dailyAnalysisCount	integer	Resets daily at midnight UTC
dailyAnalysisResetAt	timestamp	Reset anchor timestamp
createdAt	timestamp	Auto-generated

9.2 Analyses Table

Field	Type	Notes
id	UUID	Primary key
userId	UUID	Foreign key → users
imageUrl	string	S3 signed URL
trend	enum	bullish / bearish / sideways
trendDescription	string	AI-generated explanation

Field	Type	Notes
keyLevels	JSON	Array of { type, price, description }
technicalSignals	JSON	Array of { indicator, reading, interpretation, strength }
pointsToWatch	JSON	Array of strings
disclaimer	string	Always populated, non-empty
createdAt	timestamp	Auto-generated

10. AI Analysis Structure

The AI is prompted to return a structured JSON response. The system prompt explicitly prohibits any buy/sell language and enforces JSON-only output with no markdown.

Field	Type	Description
trend	bullish bearish sideways	Overall trend direction
trendDescription	string	Why the trend is what it is
keyLevels	array	Each entry: { type, price, description }
technicalSignals	array	Each entry: { indicator, reading, interpretation, strength }
pointsToWatch	string[]	Things to monitor before making any decision
disclaimer	string	Educational-only notice — always populated

11. Full Data Flow

Layer	Action
[Mobile App]	User picks image → validated (format + size < 10 MB) → converted to base64
[Mobile App]	POST /analysis/upload { imageBase64, mimeType } sent to API
[NestJS API]	JWT verified by JwtAuthGuard
[NestJS API]	Request body validated via Zod DTO (nestjs-zod)
[NestJS API]	Daily limit checked — ForbiddenException if free tier exceeded
[NestJS API]	Image buffer uploaded to S3 → signed imageUrl returned
[NestJS API]	Image sent to Claude Vision API with system prompt
[Claude API]	Returns raw JSON string (no markdown)
[NestJS API]	JSON parsed and validated against AnalysisResponseSchema (Zod)
[NestJS API]	Analysis persisted to PostgreSQL via Drizzle ORM
[NestJS API]	User dailyAnalysisCount incremented
[NestJS API]	Typed AnalysisResponse returned to client
[Mobile App]	Response re-validated with Zod (client-side guard)

Layer	Action
[Mobile App]	Stored in Zustand global state
[Mobile App]	Analysis Result Screen rendered from Zustand store

12. Security & Compliance

Area	Approach
Authentication	JWT with expiry + refresh token rotation
Image storage	Private S3 bucket; signed URLs only — never public
Rate limiting	Per-user throttling via NestJS throttler module
Input validation	Zod schemas enforced on both client and server
API security	Helmet headers applied on all routes
Data privacy	Uploaded images are not used for AI model training

13. Regulatory Risk Management

Risk	Mitigation
Classified as financial advice	Disclaimer on every screen + explicit T&C; acknowledged at registration
User losses attributed to app	'Not financial advice' clearly stated everywhere; no profit promises
Users interpret output as recommendations	Wording is always 'what the chart shows' — never 'what to do'
App store rejection	Framed as an educational tool throughout all store listings and metadata

14. Business Model

Plan	Price	Limits & Features
Free	\$0 / month	3 analyses per day · No export
Pro	\$19 / month	Unlimited analyses · History · PDF export
Team	\$49 / month	Up to 5 users · Shared history · PDF export

15. KPIs

Metric	Target	Notes
Activation (chart upload in session 1)	≥ 60%	Measures onboarding effectiveness
D7 Retention	≥ 30%	Week-1 stickiness benchmark
Free → Pro Conversion	≥ 8%	Revenue conversion benchmark

Metric	Target	Notes
NPS	≥ 50	Net Promoter Score
Avg analyses per active user / week	≥ 5	Engagement depth signal

16. MVP Timeline

Week	Deliverable
Week 1	NX Monorepo setup · shared types · ESLint + TS strict config
Week 2	NestJS boilerplate · Auth module · Drizzle schema · PostgreSQL
Week 3	React Native skeleton · NativeWind setup · Navigation structure
Week 4	Image upload flow — mobile → S3 → API
Week 5	Claude Vision integration · prompt engineering · response parsing
Week 6	Analysis result UI · React Query integration · Zod response validation
Week 7	Disclaimer · T&C; · Stripe payments · tier enforcement
Week 8	Closed beta · user feedback · bug fixes

17. Open Questions

- Should we support video clips or animated GIFs of charts in Phase 2?
- Do we want to offer a web version (Next.js) alongside the mobile app?
- What is the data retention policy for uploaded chart images on S3?
- Should the AI detect and state the timeframe of the chart, or ask the user to specify it?
- What analytics platform do we use to track activation and retention KPIs?